

Week 1 - Day 3a Supplemental: While Loops

This section contains extra material related to while loops.

Exercise 1: More While Loops []

1.1 Write a program that prints the first ten integers.

```
# TODO: Your code here
```

1.2 Write a program to print the series 2, 4, 8, 16, ..., 1024

```
# TODO: Your code here
```

1.3 Find the sum of the elements of the list `lst = [10, 2, 3, 483, 4738]`.

```
# TODO: Your code here
```

1.4 Find the sum of the digits of a number `x = 19839837287`.

```
# TODO: Your code here
```

1.5 Reverse the digits of a number `x = 19839837287`

```
# TODO: Your code here
```

Exercise 2: Helping a Billionaire []

Abebe is a billionaire. As a billionaire he doesn't have much time to spend on reading things. He wants to look at his bank account and know how much money he has right away. Unfortunately, the bank's mobile app doesn't use commas to separate digits so we have to help him. Currently, the bank's mobile applicaiton says he has `M = 1123456789` amount of money. Can you write a few lines of code to make the amount of money to look like `1,123,456,789`?

Use `while` loop to make your code work for any positive amount of money (in other words, assume that `M > 0`).

Hint: When should your while loop stop?

```
# TODO: Your code here
```

Exercise 3: Product Checking []

When computing `1 * 2 * 3 * 4 * ...`, how many numbers do you need to multiply so that the product exceeds 1000?

Hint: Use a while loop.

```
# TODO: Your code here
```

Package Lesson: Random Numbers

`random.randrange(a, b)` is a function to generate a random integer between `a` and `b` (excluding `b` itself, like `range`).

To use this function, you first need to import the `random` module by running `import random` in your notebook. Run the code below to generate a random number between 1 and (excluding) 4.

```
import random  
  
random.randrange(1, 4)
```

Exercise 4: Random Number Loops []

4.1 Add up random numbers between 1 and 10 (exclusive) until their sum exceeds 1000! Count how many numbers you need and print the result.

Extension: Use a while loop to run this code many times and compute the average!

```
# TODO: Your code here
```

4.2 Generate random numbers between 1 and 10 (exclusive) until you get the number 9. Count how many numbers you need (for example, if the first number is a 9, the count is 0) and print the result.

Extension: Use a while loop to run this code many times and compute the average!

```
# TODO: Your code here
```

Exercise 5: Matrix Printing []

Given the following 5 x 5 matrix, print all elements of the matrix in spiral order.

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 10 & 9 & 8 & 7 & 6 \\ 11 & 12 & 13 & 14 & 15 \\ 20 & 19 & 18 & 17 & 16 \\ 21 & 22 & 23 & 24 & 25 \end{bmatrix}$$

The order of print should look like 1 2 3 4 5 6 7 8 ... 25

```
# TODO: Your code here
```

Exercise 6: Guessing Game []

Create a program that generates a random number between 1 and 20 and asks the user to guess it. The program should provide hints such as "Too high" or "Too low" and continue to ask for guesses until the user correctly guesses the number.

```
# TODO: Your code here
```